Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

CRITICAL CARE

Management of Severe Respiratory Failure/ARDS

YG CG 1017 July 2022

FIRST APPROVED: JULY 2022 REVIEW BY: JULY 2025

Author(s): Eirlys Walker
Reviewer: Critical Care Group

Key Points

- CG based on the Wales Critical care and Trauma network Severe Respiratory failure/ARDS Bundle (updated 2019)
- ARDS is diagnosed according to the Berlin Criteria
- ARDS is classified as Mild, Moderate or Severe using the P/F ratio
- Mortality at 28 days is approx. 20-40%

Diagnosis:

ARDS diagnosis is based on clinical criteria rather than a pathological diagnosis. Thus, all critical care patients should be considered as potentially developing ARDS.

ARDS is defined according to the Berlin Criteria. There are 4 components & must be **acute** in onset.

Timing	Within 1 week of a known clinical insult or new/worsening respiratory symptoms
Chest imaging	Bilateral opacities – not fully explained by effusions, lobar/lung collapse, or nodules
Origin of oedema	Respiratory failure not fully explained by cardiac failure or fluid overload;
	Need objective assessment (e.g. echocardiography) to exclude hydrostatic oedema if no risk factor present

Causes:

DIRECT PULMONARY	INDIRECT
Aspiration Toxic inhalation Pulmonary contusion Pneumonia Oxygen toxicity Transthoracic radiation Post lung transplantation reperfusion injury	Sepsis syndrome Severe non-thoracic trauma Hypertransfusion Burns Cardiopulmonary bypass Hypothermia/Hyperthermia Eclampsia Acute pancreatitis Embolism Disseminated intravascular coagulopathy Shock (any form) Drug overdose

$$P/F\ Ratio = \frac{PaO2}{FiO2}$$

Scoring:

ARDS severity is scored according to the patient's P/F ratio.

ARDS Severity	P/F ratio* (on PEEP +5)	Mortality
Mild	200 – 300 mmHg (27-40 kPa)	27%
Moderate	100 – 200 mmHg (13-27kPa)	32%
Severe	< 100 mmHg (<13 kPa)	45%

Management:

Bundle Element 1: Define, diagnose and investigate	Bundle Element 2: Provide general supportive measures	Bundle Element 3: Optimise non- ventilatory management	Element 4: Optimise ventilatory management
Prompt diagnosis **Measure and record P/F Ratios on all patients** Assess criteria for ARDS if P/F <26.7kPa: - Acute onset from a recognised etiology - Bilateral infiltrates on a CXR - The absence of heart failure Record in Notes Perform NBL on all ventilated admissions to critical care. Consider BAL where indicated. Consider C.O. monitoring as indicated Consider risk factors	Comply with all bundles: - sepsis (inc antivirals) - nutrition - central line and - others as appropriate, for example tracheostomy Aim for an even or a negative fluid balance after the initial fluid resuscitation. If patient develops signs of acute renal failure consider early Renal Replacement Therapy.	Consider the use of paralysing agents in the first 48 hours, whilst utilising ventilator management strategies. Review daily thereafter. Consider **prone therapy** for ARDS patients with no improvement in gas exchange Consider the use of recruitment manoeuvres where applicable and safe to do so Consider the use rotational therapy for patients who do not tolerate manual turning or are too obese to turn safely. Provide physiotherapy, as tolerated, to enhance the removal of secretions and to improve gas exchange. Develop a rehabilitation plan whilst in Critical Care and commence rehab as soon as is practicable; follow rehab guidance.	Calculate IBW Utilise lung protective strategies – aim for: PaO2 =>8kPa or SpO2>88% Avoid alveolar overdistension – **Vt = 6mls/IBW** Limit plateau pressure - **<30 cms H2O**

Management of Severe Resp Failure.

Prone position in adult critical care 2019.pdf – link for guidance on proning inc LocSSIP